

Probability
Chapter 3 - Module 2

Conditional Expectation

Richard Redner

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Theorem. As expressed in the book and most texts

$$E[X] = E[E[X \mid Y]]$$

or to be more clear about the meaning

$$E_X[X] = E_Y[E_X[X \mid Y]].$$

Proof.

In the discrete case show that

$$E[X] = \sum_y E_X[X \mid Y = y] P(Y = y)$$

and in the continuous case show that

$$E[X] = \int_{-\infty}^{\infty} E_X[X \mid Y = y] f_Y(y) dy.$$

The book proves this result in the discrete case. Here we give the proof for the continuous case.

So let $f(x, y)$ be the joint pdf for (X, Y) and let $f_Y(y)$ be the marginal density for Y .

For the sake of this argument assume that $f_Y(y) > 0$ on $[a, b]$ including the possibility that $a = -\infty$ and $b = \infty$ and that $f_Y(y)$ is zero elsewhere.

$$\begin{aligned}
\int_{-\infty}^{\infty} E_X[X \mid Y = y] f_Y(y) dy &= \int_{-\infty}^{\infty} \left(\int_a^b x f_{X|Y}(x \mid y) dx \right) f_Y(y) dy \\
&= \int_{-\infty}^{\infty} \int_a^b x \frac{f(x, y)}{f_Y(y)} f_Y(y) dx dy \\
&= \int_{-\infty}^{\infty} \int_a^b x f(x, y) dx dy \\
&= \int_a^b x \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx \\
&= \int_a^b x f_X(x) dx \\
&= E_X[X].
\end{aligned}$$

Example 1. Suppose that the distribution for the weights W for women is $N(\mu_1, \sigma_1^2)$ and that the weights for men is $N(\mu_2, \sigma_2^2)$. And also assume that in a particular population that 55% of the population are women and that 45% of the population are men.

Suppose that we choose a random person and determine their weight W .

Question: What is $E[W]$?

Answer:

Let w denote the fact that the measurement was taken from a women and m denote the fact that the measurement was taken from a man. Then

$$E[W] = E[W \mid w] P[w] + E[W \mid m] P[m] = .55\mu_1 + .45\mu_2.$$

In Example 1. it is worth noting that the joint pdf for W has the form

$$p(x) = \alpha_1 p_1(x) + \alpha_2 p_2(x).$$

This is call a mixture density. Redoing our calcuations in terms of X we see that

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x p(x) dx \\ &= \int_{-\infty}^{\infty} x (\alpha_1 p_1(x) + \alpha_2 p_2(x)) dx \\ &= \alpha_1 \int_{-\infty}^{\infty} x p_1(x) dx + \alpha_2 \int_{-\infty}^{\infty} x p_2(x) dx \\ &= P(\text{class}_1) E[X \mid \text{class}_1] + P(\text{class}_2) E[X \mid \text{class}_2]. \end{aligned}$$

We generalize this further to m classes.

So suppose that we have m distinct populations with pdf's $p_1(x), p_2(x), \dots, p_m(x)$ and that the probability of a sample coming from each subpopulation is $\alpha_1, \alpha_2, \dots, \alpha_m$, where $\sum_{i=1}^m \alpha_i = 1$ and $\alpha_i \geq 0$ for $i = 1, 2, \dots, m$.

If X is a random sample from a random class, the pdf is

$$p(x) = \sum_{i=1}^m \alpha_i p_i(x).$$

Example 2 (3.10). In this example we compute the expected value of a random number of random variables. This is called a compound random variable.

For example, suppose that an insurance company estimates that the cost of each accident is \$8,000. And suppose that the number of accidents over a given period of time is a random number N with mean 50. What is the expected total cost.

To answer this question, let $X_1, X_2, \dots$ be random variables with mean $\mu = E[X_i]$ and for convenience let X also be a random variable with mean $\mu = E[X]$.

Define $Y = \sum_{i=1}^N X_i$.

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Solution. We begin by noting that

$$\begin{aligned} E \left[\sum_{i=1}^N X_i \mid N = n \right] &= E \left[\sum_{i=1}^n X_i \mid N = n \right] \\ &= E \left[\sum_{i=1}^n X_i \right] \\ &= n E[X]. \end{aligned}$$

So

$$E \left[\sum_{i=1}^N X_i \mid N \right] = N E[X].$$

And therefore

$$E[X] = E \left[E \left[\sum_{i=1}^N X_i \mid N \right] \right] = E[N E[X]] = E[N] E[X].$$

Quicksort is a divide and conquer algorithm for sorting numbers.

The problem at hand is given n distinct values, $x_1, x_2, \dots, x_n$ to sort the values (i.e. reorder the values so they are say in increasing order for example).

A naive choice is to choose the smallest value, then choose the next smallest value etc.

But for a list of length n , the number of comparisons is

$$(n-1) + (n-2) + \dots + 3 + 2 + 1 = \frac{n(n-1)}{2}.$$

So this naive scheme is quite a bit of work of work when n is large. There are many many schemes that are much more efficient.

Description of Quicksort.

When $n = 1$, the list is already sorted.

When $n = 2$, we simply compare the two values and reorder them as necessary.

For $n > 2$, randomly select a value, say x_i , and compare x_i to all of the values in the list and group them into two sets:

S_i , which contains all values less than x_i

and

$\bar{S}_i$, which contains all values greater than x_i .

Apply these rules recursively until the list is completely sorted.

Let's work an example.

Analysis. Let M_n denote the expected number of comparisons required to sort a random list of n distinct numbers. Then

$$\begin{aligned}M_n &= \sum_{j=1}^n E[\text{no. of comparisons} \mid \text{selected the } j\text{th smallest value}] \\&\quad \times P(\text{choosing } x_j) \\&= \sum_{j=1}^n \left((n-1) + M_{j-1} + M_{n-j} \right) P(\text{choosing } x_j) \\&= \sum_{j=1}^n \left((n-1) + M_{j-1} + M_{n-j} \right) \frac{1}{n} \\&= (n-1) + \frac{1}{n} \sum_{j=1}^n M_{j-1} + \frac{1}{n} \sum_{j=1}^n M_{n-j} \\&= (n-1) + \frac{2}{n} \sum_{k=1}^{n-1} M_k,\end{aligned}$$

where we define $M_0 = 0$.

Our answer,

$$M_n = (n - 1) + \frac{2}{n} \sum_{k=1}^{n-1} M_k \quad \text{with} \quad M_0 = 0$$

for $n \geq 1$ recursively defines the solution.

To take this a little bit further note that we can write this as

$$n M_n = n(n - 1) + \sum_{k=1}^{n-1} M_k$$

and by replacing n with $n + 1$ we get.

$$(n + 1)M_{n+1} = (n + 1)n + \sum_{k=1}^n M_k.$$

With little more algebra (see the text) we get for $n \geq 1$ that

$$M_{n+1} = 2(n + 2) \sum_{i=1}^n \frac{i}{(i + 1)(i + 2)} \approx 2(n + 2) \log(n + 2).$$